

# Solar Fe I appendix

Sun · Fe I · downloadable appendix

Fe.json v1.221 · 130 Fe I products · 160 total

Solar Fe I anchor:  $7.477 \pm 0.052$

HARPS · Reference Grade · Amarsi 3D-NLTE ·  $n = 21$ . The selected Fe I product anchors this Solar appendix; products and engines are not averaged together.

## Highlighted band products

### near-UV

**$7.565 \pm 0.129$**

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade ·  $n = 56$

Systematic under development — the abundance is current (post-RYA-1207 molecular opacity) but the uncertainty budget is not closed: component wiring and the blends bound are pending RYA-1226.

### VIS

**$7.504 \pm 0.049$**

Kitt Peak Molecfit · Synth · 3D-NLTE · Amarsi · Reference Grade ·  $n = 21$

### VIS

**$7.477 \pm 0.052$**

HARPS · Synth · 3D-NLTE · Amarsi · Reference Grade ·  $n = 21$

### red-optical

**$7.498 \pm 0.059$**

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade ·  $n = 69$

### NIR

**$7.487 \pm 0.087$**

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade ·  $n = 6$

### H

**$7.523 \pm 0.092$**

CRIRES+ · Synth · 1D-LTE · Gerber · Codex Grade ·  $n = 19$

## Error-bar forest

**How to read this forest.** Each measured row is a separate abundance product for an instrument, holding, line pool and treatment; engines are never averaged together. Solid bars show statistical uncertainty from the line measurements; wireframes show systematic uncertainty from the product's adopted error budget, which does not average away with more lines. Green and gold regions mark the Asplund and Lodders literature values and uncertainties, not pass/fail gates. Grades describe the line pool's provenance and selection, not the quality of an engine. Reddish-orange mean-3D / Frankenstein rows are experimental and not adopted. Unmeasured model rows remain N/A.

Grades and model labels

**Reference Grade.** An externally published reference line pool, using that reference's line selection and oscillator strengths; the current Fe reference products use AGSS21. This is a provenance category, not a promise that every line has an individual primary-laboratory uncertainty.

**Codex Grade.** The vetted Codex pool of primary-laboratory oscillator strengths, restricted to lines at or below the adopted feature-depth gate.

**Deep Grade.** The corresponding laboratory-data pool above that depth gate. These stronger lines extend coverage but require particular care with saturation and blends.

**1D-LTE** uses a one-dimensional atmosphere with atomic populations in local thermodynamic equilibrium. **1D-NLTE** allows departures from that equilibrium; the named implementation supplies the correction or synthesis treatment.

**3D-NLTE (Amarsi)** includes three-dimensional atmospheric structure and non-LTE line formation; these products apply the published Amarsi correction grid to a 1D baseline.

**Mean-3D / Frankenstein** combines a spatially and temporally averaged 3D atmosphere with the Gerber synthesis route. Averaging removes the full 3D fluctuations and velocity field; this experimental hybrid is not a full 3D calculation or an adopted engine.

**Synthesis** fits the spectral line profile. **EW** infers abundance from a line's integrated absorption (equivalent width). See [Methodology](#) for the full model, line-selection and uncertainty procedures.

Asplund, Amarsi & Grevesse 2021  $7.46 \pm 0.04$  · Lodders+ 2025  $7.49 \pm 0.01$

blue solid = statistical  $\sigma$  · wireframe = systematic  $\sigma$  · blue = regular model result · reddish orange = experimental Frankenstein / Gerber mean-3D; not adopted · green = Asplund; gold = Lodders · N/A = no product for this model

### near-UV — Kitt Peak — Kurucz 2005 corrected solar\_kpno\_kurucz2005\_corrected · DEEPGRADED (no graded product in this band) A(Fe) dex; axis 7.36-7.71

|                                                    |                                                                                      |                                  |
|----------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------|
| EW · 1D-LTE<br>no product                          |                                                                                      | N/A                              |
| Synth · 1D-LTE<br>Deep Grade · n=55                |  | 7.554<br>±0.061 stat ±0.116 syst |
| Synth · 1D-LTE · Gerber<br>no product              |                                                                                      | N/A                              |
| EW · 1D-NLTE · Bergemann<br>no product             |                                                                                      | N/A                              |
| Synth · 1D-NLTE · Bergemann<br>Deep Grade · n=55   |  | 7.562<br>±0.061 stat ±0.116 syst |
| Synth · 1D-NLTE · Gerber<br>no product             |                                                                                      | N/A                              |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |                                                                                      | N/A                              |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |                                                                                      | N/A                              |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |                                                                                      | N/A                              |

### near-UV — Kitt Peak — Kurucz 2005 corrected solar\_kpno\_kurucz2005\_corrected A(Fe) dex; axis 7.36-7.71

|                              |  |     |
|------------------------------|--|-----|
| EW · 1D-LTE<br>no product    |  | N/A |
| Synth · 1D-LTE<br>no product |  | N/A |

## near-UV — Kitt Peak — Kurucz 2005 corrected solar\_kpno\_kurucz2005\_corrected A(Fe) dex; axis 7.36-7.71

|                                       |     |
|---------------------------------------|-----|
| Synth · 1D-LTE · Gerber<br>no product | N/A |
|---------------------------------------|-----|

|                                        |     |
|----------------------------------------|-----|
| EW · 1D-NLTE · Bergemann<br>no product | N/A |
|----------------------------------------|-----|

|                                           |     |
|-------------------------------------------|-----|
| Synth · 1D-NLTE · Bergemann<br>no product | N/A |
|-------------------------------------------|-----|

|                                        |     |
|----------------------------------------|-----|
| Synth · 1D-NLTE · Gerber<br>no product | N/A |
|----------------------------------------|-----|

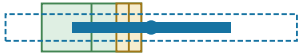
|                                                   |     |
|---------------------------------------------------|-----|
| Synth · <3D>-LTE · Gerber · stagger<br>no product | N/A |
|---------------------------------------------------|-----|

|                                                    |     |
|----------------------------------------------------|-----|
| Synth · <3D>-NLTE · Gerber · stagger<br>no product | N/A |
|----------------------------------------------------|-----|

|                                                 |     |
|-------------------------------------------------|-----|
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product | N/A |
|-------------------------------------------------|-----|

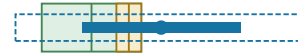
## near-UV — Kitt Peak — 1984 Molecfit corrected solar\_kpno\_molecfit\_corrected · DEEPGADED (no graded product in this band) A(Fe) dex; axis 7.36-7.71

|                           |     |
|---------------------------|-----|
| EW · 1D-LTE<br>no product | N/A |
|---------------------------|-----|

|                                     |                                                                                      |                                  |
|-------------------------------------|--------------------------------------------------------------------------------------|----------------------------------|
| Synth · 1D-LTE<br>Deep Grade · n=54 |  | 7.508<br>±0.064 stat ±0.117 syst |
|-------------------------------------|--------------------------------------------------------------------------------------|----------------------------------|

|                                       |     |
|---------------------------------------|-----|
| Synth · 1D-LTE · Gerber<br>no product | N/A |
|---------------------------------------|-----|

|                                        |     |
|----------------------------------------|-----|
| EW · 1D-NLTE · Bergemann<br>no product | N/A |
|----------------------------------------|-----|

|                                                  |                                                                                      |                                  |
|--------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------|
| Synth · 1D-NLTE · Bergemann<br>Deep Grade · n=54 |  | 7.516<br>±0.064 stat ±0.117 syst |
|--------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------|

|                                        |     |
|----------------------------------------|-----|
| Synth · 1D-NLTE · Gerber<br>no product | N/A |
|----------------------------------------|-----|

|                                                   |     |
|---------------------------------------------------|-----|
| Synth · <3D>-LTE · Gerber · stagger<br>no product | N/A |
|---------------------------------------------------|-----|

|                                                    |     |
|----------------------------------------------------|-----|
| Synth · <3D>-NLTE · Gerber · stagger<br>no product | N/A |
|----------------------------------------------------|-----|

|                                                 |     |
|-------------------------------------------------|-----|
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product | N/A |
|-------------------------------------------------|-----|

## near-UV — Kitt Peak — 1984 Molecfit corrected solar\_kpno\_molecfit\_corrected A(Fe) dex; axis 7.36-7.71

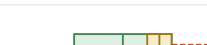
|                           |     |
|---------------------------|-----|
| EW · 1D-LTE<br>no product | N/A |
|---------------------------|-----|

## near-UV — Kitt Peak — 1984 Molecfit corrected solar\_kpno\_molecfit\_corrected A(Fe) dex; axis 7.36-7.71

|                                                    |     |
|----------------------------------------------------|-----|
| Synth · 1D-LTE<br>no product                       | N/A |
| Synth · 1D-LTE · Gerber<br>no product              | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    | N/A |

near-UV axis: 7.36 · 7.45 · 7.53 · 7.62 · 7.71 A(Fe) dex

## VIS — Kitt Peak — Kurucz 2005 corrected solar\_kpno\_kurucz2005\_corrected A(Fe) dex; axis 7.31-7.66

|                                                                                             |                                                                                     |                                  |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------|
| EW · 1D-LTE<br>Codex Grade · n=13 · 1 alternate                                             |  | 7.445<br>±0.062 stat ±0.083 syst |
| Synth · 1D-LTE<br>Codex Grade · n=67                                                        |  | 7.442<br>±0.019 stat ±0.081 syst |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=67                                               |  | 7.439<br>±0.021 stat ±0.063 syst |
| EW · 1D-NLTE · Bergemann<br>Codex Grade · n=12                                              |  | 7.455<br>±0.067 stat ±0.084 syst |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=61                                           |  | 7.448<br>±0.019 stat ±0.081 syst |
| Synth · 1D-NLTE · Gerber<br>Codex Grade · n=67                                              |  | 7.488<br>±0.020 stat ±0.063 syst |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein<br>(experimental)<br>Codex Grade · n=67  |  | 7.541<br>±0.020 stat ±0.063 syst |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein<br>(experimental)<br>Codex Grade · n=67 |  | 7.562<br>±0.020 stat ±0.063 syst |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>Codex Grade · n=50                                     |  | 7.505<br>±0.022 stat ±0.058 syst |

## VIS — Kitt Peak — Kurucz 2005 corrected solar\_kpno\_kurucz2005\_corrected A(Fe) dex; axis 7.31-7.66

|                                                    |     |
|----------------------------------------------------|-----|
| EW · 1D-LTE<br>no product                          | N/A |
| Synth · 1D-LTE<br>no product                       | N/A |
| Synth · 1D-LTE · Gerber<br>no product              | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    | N/A |

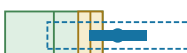
## VIS — Kitt Peak — 1984 Molecfit corrected solar\_kpno\_molecfit\_corrected A(Fe) dex; axis 7.31-7.66

|                                                                                             |                                                                                      |                                  |
|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------|
| EW · 1D-LTE<br>Codex Grade · n=13                                                           |   | 7.463<br>±0.062 stat ±0.092 syst |
| Synth · 1D-LTE<br>Codex Grade · n=67                                                        |   | 7.447<br>±0.020 stat ±0.081 syst |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=67                                               |   | 7.451<br>±0.021 stat ±0.063 syst |
| EW · 1D-NLTE · Bergemann<br>Codex Grade · n=12                                              |   | 7.464<br>±0.066 stat ±0.091 syst |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=61                                           |   | 7.454<br>±0.020 stat ±0.082 syst |
| Synth · 1D-NLTE · Gerber<br>Codex Grade · n=67                                              |   | 7.497<br>±0.022 stat ±0.063 syst |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein<br>(experimental)<br>Codex Grade · n=67  |  | 7.552<br>±0.022 stat ±0.063 syst |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein<br>(experimental)<br>Codex Grade · n=67 |  | 7.552<br>±0.022 stat ±0.063 syst |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>Codex Grade · n=50                                     |  | 7.512<br>±0.022 stat ±0.058 syst |

## VIS — Kitt Peak — 1984 Molecfit corrected solar\_kpno\_molecfit\_corrected A(Fe) dex; axis 7.31-7.66

|                                                    |     |
|----------------------------------------------------|-----|
| EW · 1D-LTE<br>no product                          | N/A |
| Synth · 1D-LTE<br>no product                       | N/A |
| Synth · 1D-LTE · Gerber<br>no product              | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    | N/A |

## VIS — HARPS solar\_harps\_molecfit\_corrected A(Fe) dex; axis 7.31-7.66

|                                                         |                                                                                      |                                  |
|---------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------|
| EW · 1D-LTE<br>Codex Grade · n=6 · 1 alternate          |  | 7.498<br>±0.070 stat ±0.097 syst |
| Synth · 1D-LTE<br>Codex Grade · n=67                    |  | 7.511<br>±0.020 stat ±0.083 syst |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=67           |  | 7.509<br>±0.021 stat ±0.083 syst |
| EW · 1D-NLTE · Bergemann<br>Codex Grade · n=5           |   | 7.433<br>±0.077 stat ±0.095 syst |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=61       |  | 7.504<br>±0.020 stat ±0.083 syst |
| Synth · 1D-NLTE · Gerber<br>Codex Grade · n=67          |  | 7.545<br>±0.020 stat ±0.083 syst |
| Synth · <3D>-LTE · Gerber · stagger<br>no product       |                                                                                      | N/A                              |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product      |                                                                                      | N/A                              |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>Codex Grade · n=50 |  | 7.512<br>±0.024 stat ±0.058 syst |

## VIS — HARPS solar\_harps\_molecfite\_corrected A(Fe) dex; axis 7.31-7.66

|                                                    |     |
|----------------------------------------------------|-----|
| EW · 1D-LTE<br>no product                          | N/A |
| Synth · 1D-LTE<br>no product                       | N/A |
| Synth · 1D-LTE · Gerber<br>no product              | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    | N/A |

## VIS — IAG solar\_iag A(Fe) dex; axis 7.31-7.66

|                                                         |                                                                                      |                                  |
|---------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------|
| EW · 1D-LTE<br>Codex Grade · n=5                        |  | 7.498<br>±0.044 stat ±0.093 syst |
| Synth · 1D-LTE<br>Codex Grade · n=43                    |  | 7.494<br>±0.023 stat ±0.080 syst |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=43           |  | 7.492<br>±0.023 stat ±0.087 syst |
| EW · 1D-NLTE · Bergemann<br>Codex Grade · n=4           |  | 7.481<br>±0.019 stat ±0.089 syst |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=37       |   | 7.480<br>±0.023 stat ±0.080 syst |
| Synth · 1D-NLTE · Gerber<br>Codex Grade · n=43          |  | 7.549<br>±0.023 stat ±0.083 syst |
| Synth · <3D>-LTE · Gerber · stagger<br>no product       |                                                                                      | N/A                              |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product      |                                                                                      | N/A                              |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>Codex Grade · n=39 |  | 7.543<br>±0.025 stat ±0.056 syst |

## VIS — IAG solar\_iag A(Fe) dex; axis 7.31–7.66

|                                                    |  |     |
|----------------------------------------------------|--|-----|
| EW · 1D-LTE<br>no product                          |  | N/A |
| Synth · 1D-LTE<br>no product                       |  | N/A |
| Synth · 1D-LTE · Gerber<br>no product              |  | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             |  | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          |  | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             |  | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |  | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |  | N/A |

VIS axis: 7.31 · 7.40 · 7.49 · 7.57 · 7.66 A(Fe) dex

## red-optical — Kitt Peak — Kurucz 2005 corrected solar\_kpno\_kurucz2005\_corrected A(Fe) dex; axis 7.35–7.59

|                                                    |                                                                                      |                                  |
|----------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------|
| EW · 1D-LTE<br>no product                          |                                                                                      | N/A                              |
| Synth · 1D-LTE<br>Codex Grade · n=69               |  | 7.458<br>±0.017 stat ±0.069 syst |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=69      |   | 7.454<br>±0.016 stat ±0.057 syst |
| EW · 1D-NLTE · Bergemann<br>no product             |                                                                                      | N/A                              |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=35  |   | 7.451<br>±0.017 stat ±0.071 syst |
| Synth · 1D-NLTE · Gerber<br>Codex Grade · n=69     |  | 7.498<br>±0.016 stat ±0.057 syst |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |                                                                                      | N/A                              |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |                                                                                      | N/A                              |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |                                                                                      | N/A                              |

## red-optical — Kitt Peak — Kurucz 2005 corrected solar\_kpno\_kurucz2005\_corrected A(Fe) dex; axis 7.35-7.59

|                                                    |     |
|----------------------------------------------------|-----|
| EW · 1D-LTE<br>no product                          | N/A |
| Synth · 1D-LTE<br>no product                       | N/A |
| Synth · 1D-LTE · Gerber<br>no product              | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    | N/A |

## red-optical — Kitt Peak — 1984 Molecfit corrected solar\_kpno\_molecfit\_corrected A(Fe) dex; axis 7.35-7.59

|                                                    |                                                                                      |                                  |
|----------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------|
| EW · 1D-LTE<br>no product                          |                                                                                      | N/A                              |
| Synth · 1D-LTE<br>Codex Grade · n=53               |  | 7.484<br>±0.029 stat ±0.070 syst |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=53      |  | 7.478<br>±0.028 stat ±0.055 syst |
| EW · 1D-NLTE · Bergemann<br>no product             |                                                                                      | N/A                              |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=27  |  | 7.493<br>±0.025 stat ±0.070 syst |
| Synth · 1D-NLTE · Gerber<br>Codex Grade · n=53     |  | 7.507<br>±0.029 stat ±0.055 syst |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |                                                                                      | N/A                              |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |                                                                                      | N/A                              |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |                                                                                      | N/A                              |

## red-optical — Kitt Peak — 1984 Molecfiit corrected solar\_kpno\_molecfiit\_corrected A(Fe) dex; axis 7.35-7.59

|                                                    |     |
|----------------------------------------------------|-----|
| EW · 1D-LTE<br>no product                          | N/A |
| Synth · 1D-LTE<br>no product                       | N/A |
| Synth · 1D-LTE · Gerber<br>no product              | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    | N/A |

## red-optical — IAG solar\_iag A(Fe) dex; axis 7.35-7.59

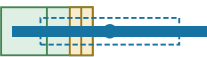
|                                                    |                                  |
|----------------------------------------------------|----------------------------------|
| EW · 1D-LTE<br>no product                          | N/A                              |
| Synth · 1D-LTE<br>Codex Grade · n=68               | 7.448<br>±0.024 stat ±0.070 syst |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=68      | 7.453<br>±0.024 stat ±0.070 syst |
| EW · 1D-NLTE · Bergemann<br>no product             | N/A                              |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=35  | 7.460<br>±0.041 stat ±0.074 syst |
| Synth · 1D-NLTE · Gerber<br>Codex Grade · n=68     | 7.491<br>±0.024 stat ±0.070 syst |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  | N/A                              |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product | N/A                              |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    | N/A                              |

## red-optical — IAG solar\_iag A(Fe) dex; axis 7.35-7.59

|                                                    |  |     |
|----------------------------------------------------|--|-----|
| EW · 1D-LTE<br>no product                          |  | N/A |
| Synth · 1D-LTE<br>no product                       |  | N/A |
| Synth · 1D-LTE · Gerber<br>no product              |  | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             |  | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          |  | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             |  | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |  | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |  | N/A |

red-optical axis: 7.35 · 7.41 · 7.47 · 7.53 · 7.59 A(Fe) dex

## NIR — Kitt Peak — Kurucz 2005 corrected solar\_kpno\_kurucz2005\_corrected A(Fe) dex; axis 7.36-7.73

|                                                    |                                                                                     |                                  |
|----------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------|
| EW · 1D-LTE<br>no product                          |                                                                                     | N/A                              |
| Synth · 1D-LTE<br>no product                       |                                                                                     | N/A                              |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=23      |  | 7.515<br>±0.086 stat ±0.061 syst |
| EW · 1D-NLTE · Bergemann<br>no product             |                                                                                     | N/A                              |
| Synth · 1D-NLTE · Bergemann<br>no product          |                                                                                     | N/A                              |
| Synth · 1D-NLTE · Gerber<br>no product             |                                                                                     | N/A                              |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |                                                                                     | N/A                              |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |                                                                                     | N/A                              |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |                                                                                     | N/A                              |

## NIR — Kitt Peak — Kurucz 2005 corrected solar\_kpno\_kurucz2005\_corrected A(Fe) dex; axis 7.36-7.73

|                                                    |  |     |
|----------------------------------------------------|--|-----|
| EW · 1D-LTE<br>no product                          |  | N/A |
| Synth · 1D-LTE<br>no product                       |  | N/A |
| Synth · 1D-LTE · Gerber<br>no product              |  | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             |  | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          |  | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             |  | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |  | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |  | N/A |

## NIR — Kitt Peak — 1984 Molecfit corrected solar\_kpno\_molecfit\_corrected A(Fe) dex; axis 7.36-7.73

|                                                    |                                                                                      |                                  |
|----------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------|
| EW · 1D-LTE<br>no product                          |                                                                                      | N/A                              |
| Synth · 1D-LTE<br>Codex Grade · n=23               |  | 7.607<br>±0.080 stat ±0.065 syst |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=23      |  | 7.601<br>±0.080 stat ±0.063 syst |
| EW · 1D-NLTE · Bergemann<br>no product             |                                                                                      | N/A                              |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=6   |   | 7.488<br>±0.059 stat ±0.065 syst |
| Synth · 1D-NLTE · Gerber<br>no product             |                                                                                      | N/A                              |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |                                                                                      | N/A                              |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |                                                                                      | N/A                              |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |                                                                                      | N/A                              |

## NIR — Kitt Peak — 1984 Molecfit corrected solar\_kpno\_molecfit\_corrected A(Fe) dex; axis 7.36-7.73

|                                                    |     |
|----------------------------------------------------|-----|
| EW · 1D-LTE<br>no product                          | N/A |
| Synth · 1D-LTE<br>no product                       | N/A |
| Synth · 1D-LTE · Gerber<br>no product              | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    | N/A |

## NIR — IAG solar\_iag A(Fe) dex; axis 7.36-7.73

|                                                    |                                                                                                                                 |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| EW · 1D-LTE<br>no product                          | N/A                                                                                                                             |
| Synth · 1D-LTE<br>Codex Grade · n=25               |  7.546<br>$\pm 0.159$ stat $\pm 0.063$ syst |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=22      |  7.549<br>$\pm 0.069$ stat $\pm 0.064$ syst  |
| EW · 1D-NLTE · Bergemann<br>no product             | N/A                                                                                                                             |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=6   |  7.599<br>$\pm 0.072$ stat $\pm 0.063$ syst |
| Synth · 1D-NLTE · Gerber<br>no product             | N/A                                                                                                                             |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  | N/A                                                                                                                             |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product | N/A                                                                                                                             |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    | N/A                                                                                                                             |

## NIR — IAG solar\_iag A(Fe) dex; axis 7.36-7.73

|                                                    |  |     |
|----------------------------------------------------|--|-----|
| EW · 1D-LTE<br>no product                          |  | N/A |
| Synth · 1D-LTE<br>no product                       |  | N/A |
| Synth · 1D-LTE · Gerber<br>no product              |  | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             |  | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          |  | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             |  | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |  | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |  | N/A |

## NIR — CRIRES+ solar\_crires\_plus\_y\_wide\_rya1054 A(Fe) dex; axis 7.36-7.73

|                                                    |                                                                                     |                                  |
|----------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------|
| EW · 1D-LTE<br>no product                          |                                                                                     | N/A                              |
| Synth · 1D-LTE<br>Codex Grade · n=5                |  | 7.551<br>±0.062 stat ±0.062 syst |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=5       |  | 7.553<br>±0.062 stat ±0.062 syst |
| EW · 1D-NLTE · Bergemann<br>no product             |                                                                                     | N/A                              |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=2   |  | 7.492<br>±0.070 stat ±0.061 syst |
| Synth · 1D-NLTE · Gerber<br>no product             |                                                                                     | N/A                              |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |                                                                                     | N/A                              |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |                                                                                     | N/A                              |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |                                                                                     | N/A                              |

## NIR — CRIRES+ solar\_crires\_plus\_y\_wide\_rya1054 A(Fe) dex; axis 7.36-7.73

|                                                    |  |     |
|----------------------------------------------------|--|-----|
| EW · 1D-LTE<br>no product                          |  | N/A |
| Synth · 1D-LTE<br>no product                       |  | N/A |
| Synth · 1D-LTE · Gerber<br>no product              |  | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             |  | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          |  | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             |  | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |  | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |  | N/A |

NIR axis: 7.36 · 7.45 · 7.55 · 7.64 · 7.73 A(Fe) dex

## H — CRIRES+ solar\_crires\_plus\_h\_rya1094 A(Fe) dex; axis 7.39-7.67

|                                                    |                                                                                      |                                  |
|----------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------|
| EW · 1D-LTE<br>no product                          |                                                                                      | N/A                              |
| Synth · 1D-LTE<br>Codex Grade · n=20               |  | 7.552<br>±0.087 stat ±0.079 syst |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=19      |  | 7.523<br>±0.051 stat ±0.076 syst |
| EW · 1D-NLTE · Bergemann<br>no product             |                                                                                      | N/A                              |
| Synth · 1D-NLTE · Bergemann<br>no product          |                                                                                      | N/A                              |
| Synth · 1D-NLTE · Gerber<br>Codex Grade · n=9      |  | 7.549<br>±0.070 stat ±0.073 syst |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |                                                                                      | N/A                              |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |                                                                                      | N/A                              |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |                                                                                      | N/A                              |

## H — CRIRES+ solar\_crires\_plus\_h\_rya1094 A(Fe) dex; axis 7.39-7.67

|                                                    |     |
|----------------------------------------------------|-----|
| EW · 1D-LTE<br>no product                          | N/A |
| Synth · 1D-LTE<br>no product                       | N/A |
| Synth · 1D-LTE · Gerber<br>no product              | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    | N/A |

H axis: 7.39 · 7.46 · 7.53 · 7.60 · 7.67 A(Fe) dex

Read all product values, including alternate line sets and experiments

CRIRES+ · Synth · 1D-LTE · Codex Grade · solar\_crires\_plus\_h\_rya1094 · GRADED · SYNTH · 1D-LTE:  $7.552 \pm 0.118$  reported total; n=20;  $\xi$  MEASURED

CRIRES+ · Synth · 1D-NLTE · Gerber · Codex Grade · solar\_crires\_plus\_h\_rya1094 · GRADED · SYNTH · ENGINE-B-NLTE:  $7.549 \pm 0.102$  reported total; n=9;  $\xi$  MEASURED

CRIRES+ · Synth · 1D-LTE · Gerber · Codex Grade · solar\_crires\_plus\_h\_rya1094 · GRADED · SYNTH · synth-1D-LTE-gerber:  $7.523 \pm 0.092$  reported total; n=19;  $\xi$  MEASURED

CRIRES+ · Synth · 1D-LTE · Reference Grade · solar\_crires\_plus\_h\_rya1094 · REFERENCE · SYNTH · 1D-LTE:  $7.552 \pm 0.118$  reported total; n=20;  $\xi$  MEASURED

CRIRES+ · Synth · 1D-NLTE · Gerber · Reference Grade · solar\_crires\_plus\_h\_rya1094 · REFERENCE · SYNTH · ENGINE-B-NLTE:  $7.549 \pm 0.102$  reported total; n=9;  $\xi$  MEASURED

CRIRES+ · Synth · 1D-LTE · Gerber · Reference Grade · solar\_crires\_plus\_h\_rya1094 · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.518 \pm 0.092$  reported total; n=19;  $\xi$  MEASURED

CRIRES+ · Synth · 1D-LTE · Codex Grade · solar\_crires\_plus\_y\_wide\_rya1054 · GRADED · SYNTH · 1D-LTE:  $7.551 \pm 0.087$  reported total; n=5;  $\xi$  MEASURED

CRIRES+ · Synth · 1D-NLTE · Bergemann · Codex Grade · solar\_crires\_plus\_y\_wide\_rya1054 · GRADED · SYNTH · ENGINE-A:  $7.492 \pm 0.093$  reported total; n=2;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

CRIRES+ · Synth · 1D-LTE · Gerber · Codex Grade · solar\_crires\_plus\_y\_wide\_rya1054 · GRADED · SYNTH · synth-1D-LTE-gerber:  $7.553 \pm 0.088$  reported total; n=5;  $\xi$  MEASURED

CRIRES+ · Synth · 1D-LTE · Reference Grade · solar\_crires\_plus\_y\_wide\_rya1054 · REFERENCE · SYNTH · 1D-LTE:  $7.546 \pm 0.088$  reported total; n=5;  $\xi$  MEASURED

CRIRES+ · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_crires\_plus\_y\_wide\_rya1054 · REFERENCE · SYNTH · ENGINE-A:  $7.486 \pm 0.092$  reported total; n=2;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this

pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

CRIRES+ · Synth · 1D-LTE · Gerber · Reference Grade · solar\_crires\_plus\_y\_wide\_rya1054 · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.554 \pm 0.088$  reported total; n=5;  $\xi$  MEASURED

IAG · Synth · 1D-LTE · Codex Grade · solar\_iag · GRADED · SYNTH · 1D-LTE:  $7.546 \pm 0.171$  reported total; n=25;  $\xi$  MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Codex Grade · solar\_iag · GRADED · SYNTH · ENGINE-A:  $7.599 \pm 0.095$  reported total; n=6;  $\xi$  MEASURED

IAG · Synth · 1D-LTE · Gerber · Codex Grade · solar\_iag · GRADED · SYNTH · synth-1D-LTE-gerber:  $7.549 \pm 0.094$  reported total; n=22;  $\xi$  MEASURED

IAG · Synth · 1D-LTE · Reference Grade · solar\_iag · REFERENCE · SYNTH · 1D-LTE:  $7.554 \pm 0.092$  reported total; n=22;  $\xi$  MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_iag · REFERENCE · SYNTH · ENGINE-A:  $7.599 \pm 0.097$  reported total; n=6;  $\xi$  MEASURED

IAG · Synth · 1D-LTE · Gerber · Reference Grade · solar\_iag · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.549 \pm 0.094$  reported total; n=22;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar\_kpno\_kurucz2005\_corrected · GRADED · SYNTH · synth-1D-LTE-gerber:  $7.515 \pm 0.105$  reported total; n=23;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · 1D-LTE:  $7.519 \pm 0.106$  reported total; n=23;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · ENGINE-A:  $7.390 \pm 0.187$  reported total; n=6;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.510 \pm 0.107$  reported total; n=23;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar\_kpno\_molecfit\_corrected · GRADED · SYNTH · 1D-LTE:  $7.607 \pm 0.104$  reported total; n=23;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar\_kpno\_molecfit\_corrected · GRADED · SYNTH · ENGINE-A:  $7.488 \pm 0.087$  reported total; n=6;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar\_kpno\_molecfit\_corrected · GRADED · SYNTH · synth-1D-LTE-gerber:  $7.601 \pm 0.102$  reported total; n=23;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar\_kpno\_molecfit\_corrected · REFERENCE · SYNTH · 1D-LTE:  $7.607 \pm 0.103$  reported total; n=23;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_kpno\_molecfit\_corrected · REFERENCE · SYNTH · ENGINE-A:  $7.487 \pm 0.087$  reported total; n=6;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar\_kpno\_molecfit\_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.601 \pm 0.103$  reported total; n=23;  $\xi$  MEASURED

HARPS · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar\_harps\_molecfit\_corrected · ASPLUND\_AGSS21 · SYNTH · ENGINE-A-3DNLTE:  $7.477 \pm 0.052$  reported total; n=21;  $\xi$  NOT\_APPLICABLE

HARPS · Synth · 1D-LTE · Deep Grade · solar\_harps\_molecfit\_corrected · DEEPGRADED-LOCALRENORM · SYNTH · 1D-LTE:  $7.339 \pm 0.087$  reported total; n=109;  $\xi$  MEASURED

HARPS · Synth · 1D-NLTE · Bergemann · Deep Grade · solar\_harps\_molecfit\_corrected · DEEPGRADED-LOCALRENORM · SYNTH · ENGINE-A:  $7.350 \pm 0.088$  reported total; n=105;  $\xi$  MEASURED

HARPS · Synth · 1D-LTE · Deep Grade · solar\_harps\_molecfit\_corrected · DEEPGRADED · SYNTH · 1D-LTE:  $7.535 \pm 0.093$  reported total; n=109;  $\xi$  MEASURED

HARPS · Synth · 1D-NLTE · Bergemann · Deep Grade · solar\_harps\_molecfits\_corrected · DEEPGRADED · SYNTH · ENGINE-A:  $7.554 \pm 0.094$  reported total; n=105;  $\xi$  MEASURED

HARPS · EW · 1D-LTE · Codex Grade · solar\_harps\_molecfits\_corrected · GRADED · PROFILEFIT · 1D-LTE:  $7.498 \pm 0.120$  reported total; n=6;  $\xi$  MEASURED

HARPS · EW · 1D-NLTE · Bergemann · Codex Grade · solar\_harps\_molecfits\_corrected · GRADED · PROFILEFIT · ENGINE-A:  $7.433 \pm 0.123$  reported total; n=5;  $\xi$  MEASURED

HARPS · EW · 1D-LTE · Codex Grade · solar\_harps\_molecfits\_corrected · GRADED · PROFILEFIT · ENGINE-B:  $7.431 \pm 0.122$  reported total; n=5;  $\xi$  MEASURED

HARPS · Synth · 1D-LTE · Codex Grade · solar\_harps\_molecfits\_corrected · GRADED · SYNTH · 1D-LTE:  $7.511 \pm 0.086$  reported total; n=67;  $\xi$  MEASURED

HARPS · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar\_harps\_molecfits\_corrected · GRADED · SYNTH · ENGINE-A-3DNLTE:  $7.512 \pm 0.062$  reported total; n=50;  $\xi$  NOT\_APPLICABLE

HARPS · Synth · 1D-NLTE · Bergemann · Codex Grade · solar\_harps\_molecfits\_corrected · GRADED · SYNTH · ENGINE-A:  $7.504 \pm 0.086$  reported total; n=61;  $\xi$  MEASURED

HARPS · Synth · 1D-NLTE · Gerber · Codex Grade · solar\_harps\_molecfits\_corrected · GRADED · SYNTH · ENGINE-B-NLTE:  $7.545 \pm 0.086$  reported total; n=67;  $\xi$  MEASURED

HARPS · Synth · 1D-LTE · Gerber · Codex Grade · solar\_harps\_molecfits\_corrected · GRADED · SYNTH · synth-1D-LTE-gerber:  $7.509 \pm 0.086$  reported total; n=67;  $\xi$  MEASURED

HARPS · Synth · 1D-LTE · Reference Grade · solar\_harps\_molecfits\_corrected · REFERENCE · SYNTH · 1D-LTE:  $7.514 \pm 0.084$  reported total; n=174;  $\xi$  MEASURED

HARPS · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar\_harps\_molecfits\_corrected · REFERENCE · SYNTH · ENGINE-A-3DNLTE:  $7.500 \pm 0.048$  reported total; n=100;  $\xi$  NOT\_APPLICABLE

HARPS · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_harps\_molecfits\_corrected · REFERENCE · SYNTH · ENGINE-A:  $7.525 \pm 0.084$  reported total; n=164;  $\xi$  MEASURED

HARPS · Synth · 1D-NLTE · Gerber · Reference Grade · solar\_harps\_molecfits\_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE:  $7.549 \pm 0.083$  reported total; n=175;  $\xi$  MEASURED

HARPS · Synth · 1D-LTE · Gerber · Reference Grade · solar\_harps\_molecfits\_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.521 \pm 0.089$  reported total; n=175;  $\xi$  MEASURED

IAG · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar\_iag · ASPLUND\_AGSS21 · SYNTH · ENGINE-A-3DNLTE:  $7.504 \pm 0.050$  reported total; n=21;  $\xi$  NOT\_APPLICABLE

IAG · Synth · 1D-LTE · Deep Grade · solar\_iag · DEEPGRADED · SYNTH · 1D-LTE:  $7.416 \pm 0.071$  reported total; n=41;  $\xi$  MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Deep Grade · solar\_iag · DEEPGRADED · SYNTH · ENGINE-A:  $7.422 \pm 0.070$  reported total; n=37;  $\xi$  MEASURED

IAG · EW · 1D-LTE · Codex Grade · solar\_iag · GRADED · PROFILEFIT · 1D-LTE:  $7.498 \pm 0.102$  reported total; n=5;  $\xi$  MEASURED

IAG · EW · 1D-NLTE · Bergemann · Codex Grade · solar\_iag · GRADED · PROFILEFIT · ENGINE-A:  $7.481 \pm 0.091$  reported total; n=4;  $\xi$  MEASURED

IAG · Synth · 1D-LTE · Codex Grade · solar\_iag · GRADED · SYNTH · 1D-LTE:  $7.494 \pm 0.083$  reported total; n=43;  $\xi$  MEASURED

IAG · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar\_iag · GRADED · SYNTH · ENGINE-A-3DNLTE:  $7.543 \pm 0.061$  reported total; n=39;  $\xi$  NOT\_APPLICABLE

IAG · Synth · 1D-NLTE · Bergemann · Codex Grade · solar\_iag · GRADED · SYNTH · ENGINE-A:  $7.480 \pm 0.083$  reported total; n=37;  $\xi$  MEASURED

IAG · Synth · 1D-NLTE · Gerber · Codex Grade · solar\_iag · GRADED · SYNTH · ENGINE-B-NLTE:  $7.549 \pm 0.087$  reported total; n=43;  $\xi$  MEASURED

IAG · Synth · 1D-LTE · Gerber · Codex Grade · solar\_iag · GRADED · SYNTH · synth-1D-LTE-gerber:  $7.492 \pm 0.090$  reported total; n=43;  $\xi$  MEASURED

IAG · Synth · 1D-LTE · Reference Grade · solar\_iag · REFERENCE · SYNTH · 1D-LTE:  $7.463 \pm 0.076$  reported total; n=84;  $\xi$  MEASURED

IAG · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar\_iag · REFERENCE · SYNTH · ENGINE-A-3DNLTE:  $7.516 \pm 0.047$  reported total; n=78;  $\xi$  NOT\_APPLICABLE

IAG · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_iag · REFERENCE · SYNTH · ENGINE-A:  $7.455 \pm 0.076$  reported total; n=74;  $\xi$  MEASURED

IAG · Synth · 1D-NLTE · Gerber · Reference Grade · solar\_iag · REFERENCE · SYNTH · ENGINE-B-NLTE:  $7.517 \pm 0.077$  reported total; n=84;  $\xi$  MEASURED

IAG · Synth · 1D-LTE · Gerber · Reference Grade · solar\_iag · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.469 \pm 0.078$  reported total; n=84;  $\xi$  MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar\_kpno\_kurucz2005\_corrected · ASPLUND\_AGSS21 · SYNTH · ENGINE-A-3DNLTE:  $7.471 \pm 0.049$  reported total; n=21;  $\xi$  NOT\_APPLICABLE

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar\_kpno\_kurucz2005\_corrected · DEEPGRADED · SYNTH · 1D-LTE:  $7.421 \pm 0.086$  reported total; n=108;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar\_kpno\_kurucz2005\_corrected · DEEPGRADED · SYNTH · ENGINE-A:  $7.431 \pm 0.086$  reported total; n=104;  $\xi$  MEASURED

Kitt Peak · EW · 1D-LTE · Codex Grade · solar\_kpno\_kurucz2005\_corrected · GRADED · PROFILEFIT · 1D-LTE:  $7.445 \pm 0.104$  reported total; n=13;  $\xi$  MEASURED

Kitt Peak · EW · 1D-NLTE · Bergemann · Codex Grade · solar\_kpno\_kurucz2005\_corrected · GRADED · PROFILEFIT · ENGINE-A:  $7.455 \pm 0.107$  reported total; n=12;  $\xi$  MEASURED

Kitt Peak · EW · 1D-LTE · Codex Grade · solar\_kpno\_kurucz2005\_corrected · GRADED · PROFILEFIT · ENGINE-B:  $7.479 \pm 0.090$  reported total; n=9;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar\_kpno\_kurucz2005\_corrected · GRADED · SYNTH · 1D-LTE:  $7.442 \pm 0.084$  reported total; n=67;  $\xi$  MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar\_kpno\_kurucz2005\_corrected · GRADED · SYNTH · ENGINE-A-3DNLTE:  $7.505 \pm 0.062$  reported total; n=50;  $\xi$  NOT\_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar\_kpno\_kurucz2005\_corrected · GRADED · SYNTH · ENGINE-A:  $7.448 \pm 0.084$  reported total; n=61;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar\_kpno\_kurucz2005\_corrected · GRADED · SYNTH · ENGINE-B-NLTE:  $7.488 \pm 0.066$  reported total; n=67;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar\_kpno\_kurucz2005\_corrected · GRADED · SYNTH · synth-1D-LTE-gerber:  $7.439 \pm 0.067$  reported total; n=67;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Codex Grade · solar\_kpno\_kurucz2005\_corrected · GRADED · SYNTH · synth-mean3D-LTE-gerber-stagger:  $7.541 \pm 0.067$  reported total; n=67;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Codex Grade · solar\_kpno\_kurucz2005\_corrected · GRADED · SYNTH · synth-mean3D-NLTE-gerber-stagger:  $7.562 \pm 0.066$  reported total; n=67;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · 1D-LTE:  $7.435 \pm 0.083$  reported total; n=175;  $\xi$  MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · ENGINE-A-3DNLTE:  $7.485 \pm 0.048$  reported total; n=101;  $\xi$  NOT\_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · ENGINE-A:  $7.441 \pm 0.083$  reported total; n=165;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE:  $7.481 \pm 0.080$  reported total; n=176;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.436 \pm 0.080$  reported total; n=176;  $\xi$  MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar\_kpno\_molecfit\_corrected · ASPLUND\_AGSS21 · SYNTH · ENGINE-A-3DNLTE:  $7.504 \pm 0.049$  reported total; n=21;  $\xi$  NOT\_APPLICABLE

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar\_kpno\_molecfit\_corrected · DEEPGRADED · SYNTH · 1D-LTE:  $7.417 \pm 0.045$  reported total; n=108;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar\_kpno\_molecfit\_corrected · DEEPGRADED · SYNTH · ENGINE-A:  $7.426 \pm 0.084$  reported total; n=104;  $\xi$  MEASURED

Kitt Peak · EW · 1D-LTE · Codex Grade · solar\_kpno\_molecfit\_corrected · GRADED · PROFILEFIT · 1D-LTE:  $7.463 \pm 0.111$  reported total; n=13;  $\xi$  MEASURED

Kitt Peak · EW · 1D-NLTE · Bergemann · Codex Grade · solar\_kpno\_molecfit\_corrected · GRADED · PROFILEFIT · ENGINE-A:  $7.464 \pm 0.112$  reported total; n=12;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar\_kpno\_molecfit\_corrected · GRADED · SYNTH · 1D-LTE:  $7.447 \pm 0.084$  reported total; n=67;  $\xi$  MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar\_kpno\_molecfit\_corrected · GRADED · SYNTH · ENGINE-A-3DNLTE:  $7.512 \pm 0.062$  reported total; n=50;  $\xi$  NOT\_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar\_kpno\_molecfit\_corrected · GRADED · SYNTH · ENGINE-A:  $7.454 \pm 0.085$  reported total; n=61;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar\_kpno\_molecfit\_corrected · GRADED · SYNTH · ENGINE-B-NLTE:  $7.497 \pm 0.067$  reported total; n=67;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar\_kpno\_molecfit\_corrected · GRADED · SYNTH · synth-1D-LTE-gerber:  $7.451 \pm 0.067$  reported total; n=67;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Codex Grade · solar\_kpno\_molecfit\_corrected · GRADED · SYNTH · synth-mean3D-LTE-gerber-stagger:  $7.552 \pm 0.067$  reported total; n=67;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Codex Grade · solar\_kpno\_molecfit\_corrected · GRADED · SYNTH · synth-mean3D-NLTE-gerber-stagger:  $7.552 \pm 0.067$  reported total; n=67;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar\_kpno\_molecfit\_corrected · REFERENCE · SYNTH · 1D-LTE:  $7.432 \pm 0.082$  reported total; n=175;  $\xi$  MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar\_kpno\_molecfit\_corrected · REFERENCE · SYNTH · ENGINE-A-3DNLTE:  $7.492 \pm 0.048$  reported total; n=101;  $\xi$  NOT\_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_kpno\_molecfit\_corrected · REFERENCE · SYNTH · ENGINE-A:  $7.442 \pm 0.082$  reported total; n=165;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar\_kpno\_molecfit\_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE:  $7.487 \pm 0.081$  reported total; n=176;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar\_kpno\_molecfits\_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.446 \pm 0.083$  reported total; n=176;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar\_kpno\_kurucz2005\_corrected · DEEPGRADED · SYNTH · 1D-LTE:  $7.554 \pm 0.131$  reported total; n=55;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar\_kpno\_kurucz2005\_corrected · DEEPGRADED · SYNTH · ENGINE-A:  $7.562 \pm 0.131$  reported total; n=55;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · 1D-LTE:  $7.560 \pm 0.136$  reported total; n=56;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · ENGINE-A:  $7.569 \pm 0.136$  reported total; n=56;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.565 \pm 0.129$  reported total; n=56;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar\_kpno\_molecfits\_corrected · DEEPGRADED · SYNTH · 1D-LTE:  $7.508 \pm 0.133$  reported total; n=54;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar\_kpno\_molecfits\_corrected · DEEPGRADED · SYNTH · ENGINE-A:  $7.516 \pm 0.133$  reported total; n=54;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar\_kpno\_molecfits\_corrected · REFERENCE · SYNTH · 1D-LTE:  $7.512 \pm 0.140$  reported total; n=55;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_kpno\_molecfits\_corrected · REFERENCE · SYNTH · ENGINE-A:  $7.518 \pm 0.140$  reported total; n=55;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar\_kpno\_molecfits\_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.467 \pm 0.129$  reported total; n=57;  $\xi$  MEASURED

IAG · Synth · 1D-LTE · Codex Grade · solar\_iag · GRADED · SYNTH · 1D-LTE:  $7.448 \pm 0.074$  reported total; n=68;  $\xi$  MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Codex Grade · solar\_iag · GRADED · SYNTH · ENGINE-A:  $7.460 \pm 0.084$  reported total; n=35;  $\xi$  MEASURED

IAG · Synth · 1D-NLTE · Gerber · Codex Grade · solar\_iag · GRADED · SYNTH · ENGINE-B-NLTE:  $7.491 \pm 0.074$  reported total; n=68;  $\xi$  MEASURED

IAG · Synth · 1D-LTE · Gerber · Codex Grade · solar\_iag · GRADED · SYNTH · synth-1D-LTE-gerber:  $7.453 \pm 0.074$  reported total; n=68;  $\xi$  MEASURED

IAG · Synth · 1D-LTE · Reference Grade · solar\_iag · REFERENCE · SYNTH · 1D-LTE:  $7.448 \pm 0.074$  reported total; n=68;  $\xi$  MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_iag · REFERENCE · SYNTH · ENGINE-A:  $7.460 \pm 0.084$  reported total; n=35;  $\xi$  MEASURED

IAG · Synth · 1D-NLTE · Gerber · Reference Grade · solar\_iag · REFERENCE · SYNTH · ENGINE-B-NLTE:  $7.491 \pm 0.074$  reported total; n=68;  $\xi$  MEASURED

IAG · Synth · 1D-LTE · Gerber · Reference Grade · solar\_iag · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.453 \pm 0.074$  reported total; n=68;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar\_kpno\_kurucz2005\_corrected · GRADED · SYNTH · 1D-LTE:  $7.458 \pm 0.071$  reported total; n=69;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar\_kpno\_kurucz2005\_corrected · GRADED · SYNTH · ENGINE-A:  $7.451 \pm 0.073$  reported total; n=35;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar\_kpno\_kurucz2005\_corrected · GRADED · SYNTH · ENGINE-B-NLTE:  $7.498 \pm 0.059$  reported total; n=69;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar\_kpno\_kurucz2005\_corrected · GRADED · SYNTH · synth-1D-LTE-gerber:  $7.454 \pm 0.059$  reported total; n=69;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · 1D-LTE:  $7.458 \pm 0.072$  reported total; n=69;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · ENGINE-A:  $7.451 \pm 0.075$  reported total; n=35;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE:  $7.498 \pm 0.072$  reported total; n=69;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.454 \pm 0.076$  reported total; n=69;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar\_kpno\_molecfit\_corrected · GRADED · SYNTH · 1D-LTE:  $7.484 \pm 0.075$  reported total; n=53;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar\_kpno\_molecfit\_corrected · GRADED · SYNTH · ENGINE-A:  $7.493 \pm 0.075$  reported total; n=27;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar\_kpno\_molecfit\_corrected · GRADED · SYNTH · ENGINE-B-NLTE:  $7.507 \pm 0.062$  reported total; n=53;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar\_kpno\_molecfit\_corrected · GRADED · SYNTH · synth-1D-LTE-gerber:  $7.478 \pm 0.062$  reported total; n=53;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar\_kpno\_molecfit\_corrected · REFERENCE · SYNTH · 1D-LTE:  $7.484 \pm 0.076$  reported total; n=53;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_kpno\_molecfit\_corrected · REFERENCE · SYNTH · ENGINE-A:  $7.493 \pm 0.075$  reported total; n=27;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar\_kpno\_molecfit\_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE:  $7.507 \pm 0.074$  reported total; n=53;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar\_kpno\_molecfit\_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.478 \pm 0.078$  reported total; n=53;  $\xi$  MEASURED

## Solar iron context

Fe I supplies the bulk of the neutral-iron line statistics used for the Solar abundance reference. Separate instruments, line pools and treatments test the stability of that reference; their differences remain visible.

## Near-UV opacity — report and noted

Molecular opacity INCLUDED — OH A-X, NH, CH and CN violet, 6,223 VALD lines as Turbospectrum lists with the dominant isotopologue (RYA-1207). The band previously carried none: every molecular list iSpec ships begins at 400 nm. Measured PAIRED against a molecules-off run on the same code and pool, the lever is -0.047..-0.055 dex on Fe I and -0.276..-0.332 on Fe II, with chi2r median 81.8 -> 67.4 and control windows containing no molecular line bit-identical. RYA-1209 additionally curated out Fe I 3721.272 (systematic:K07 — gf attributed to a VALD3 reference that publishes no per-line uncertainty, no NIST accuracy class, so it never met the DEEPGRADED tier's lab-gf definition); the Fe I pool is 57 candidates and the budget clears at gf rung 3 (cited lab), 55/55 GF-LAB, RMS 0.0509 dex. RESIDUAL DEFICIT: the correction still UNDER-corrects — RYA-1204 measured synth/obs above 1 on 24 of 40 windows after it — so the band remains short of opacity by an amount not yet measured on this pool, and these values are a floor rather than a converged answer. sigma\_syst is unchanged at 0.1122 because it is dominated by the 0.10 dex pseudo-continuum term, not by gf: the true continuum is never observed here. Not adopted for the headline, which rests on the VIS anchor 7.466 (RYA-161: nothing calibrated to a target).

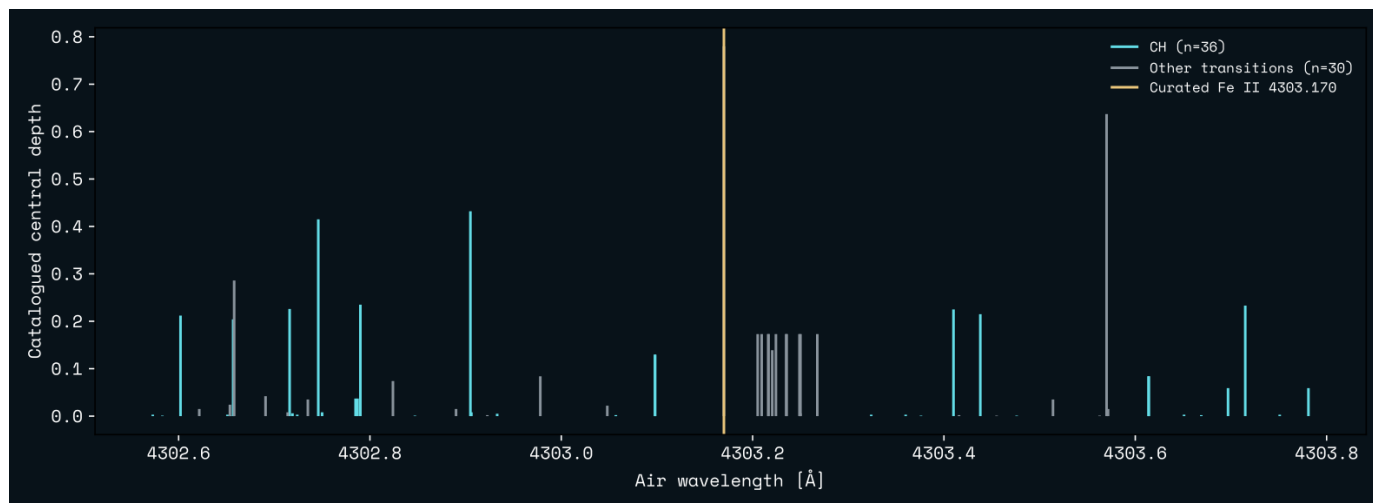
## Frankenstein — experimental, pending the Bride

4 experimental products remain available in the Fe I product values and downloads.

RYA-1197 Part A: the Frankenstein mean- $\langle 3D \rangle$  route is HELD FROM HEADLINE publication until the Bride validates it. It inflates  $A(\text{Fe})$  by roughly +0.10 dex against the 1D siblings and exists on VIS/KP only, so it is not a comparable measurement of the same quantity. It stays in the feed deliberately: the LTE/NLTE mean-3D PAIR is the NLTE differential on ONE atmosphere (RYA-542), which is the diagnostic it is for. Not a headline value.

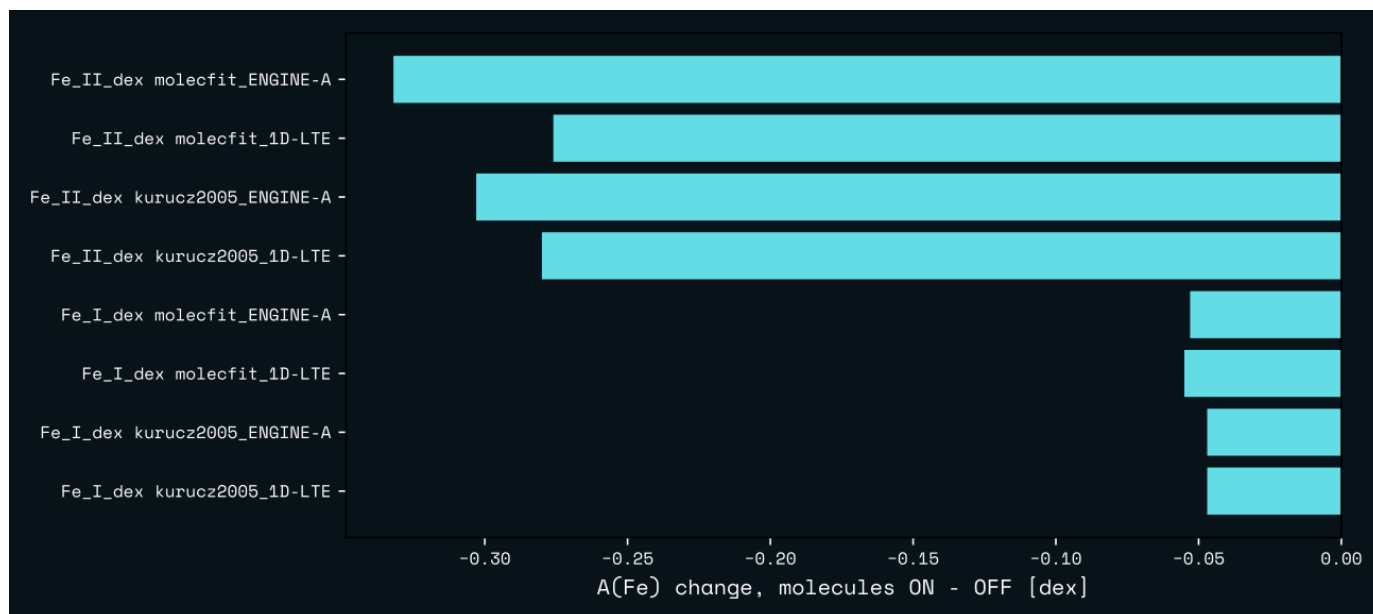
## Problem-line and model diagnostics

### CH G-band artifact



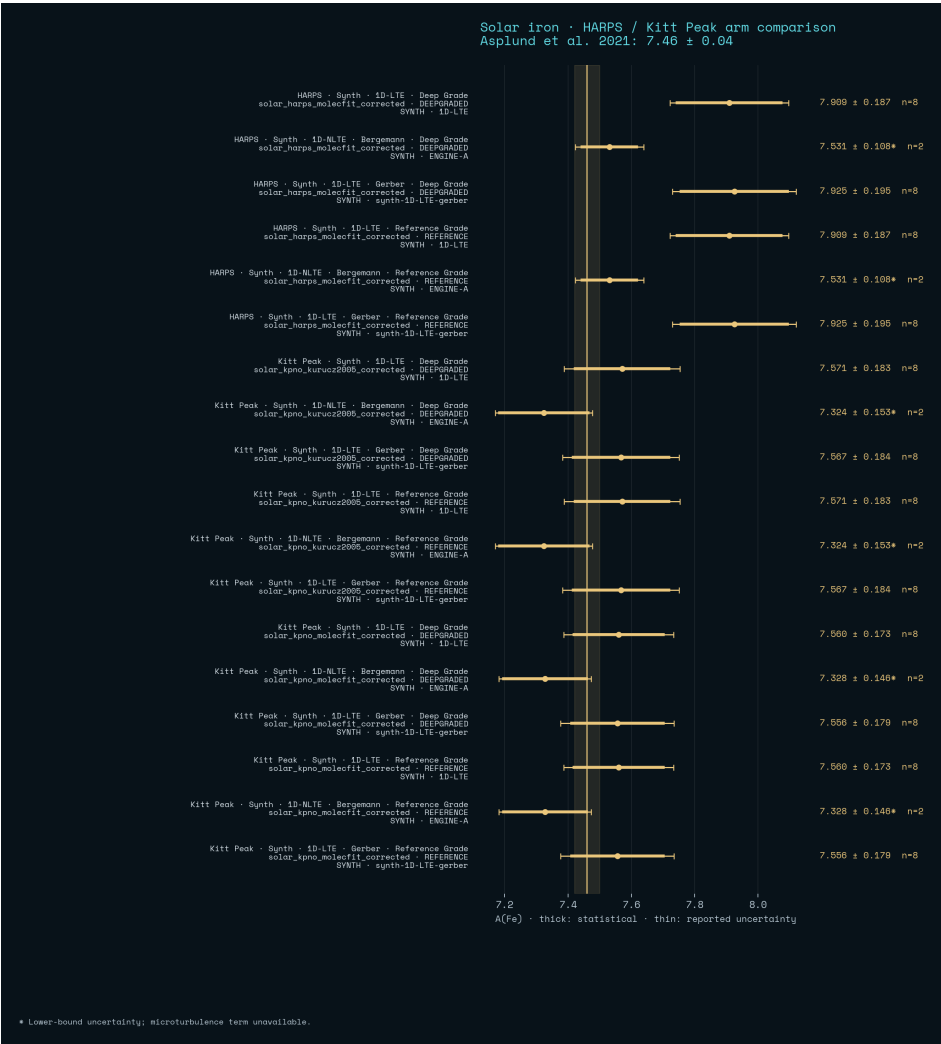
CH G-band blend artifact — the synthesis list cannot represent the dominant absorber ·  $A = 10.0$  (+2.5 dex). RYA-1191 blend audit: 66 catalogued transitions in the 0.62 Å fit window, top contaminant CH I at 43.8% of the summed depth against Fe II's 11.1%. The VIS synthesis list GESv6\_atom\_hfs\_iso carries 206 species and NOT ONE molecular, so the fit had nowhere to put the CH absorption except Fe II. RYA-515 independently counted 36 of 65 transitions as CH. The plot shows the committed transition census in the proof generator's fit window; it is not a synthetic spectrum. [Source evidence](#)

### Near-UV molecular opacity



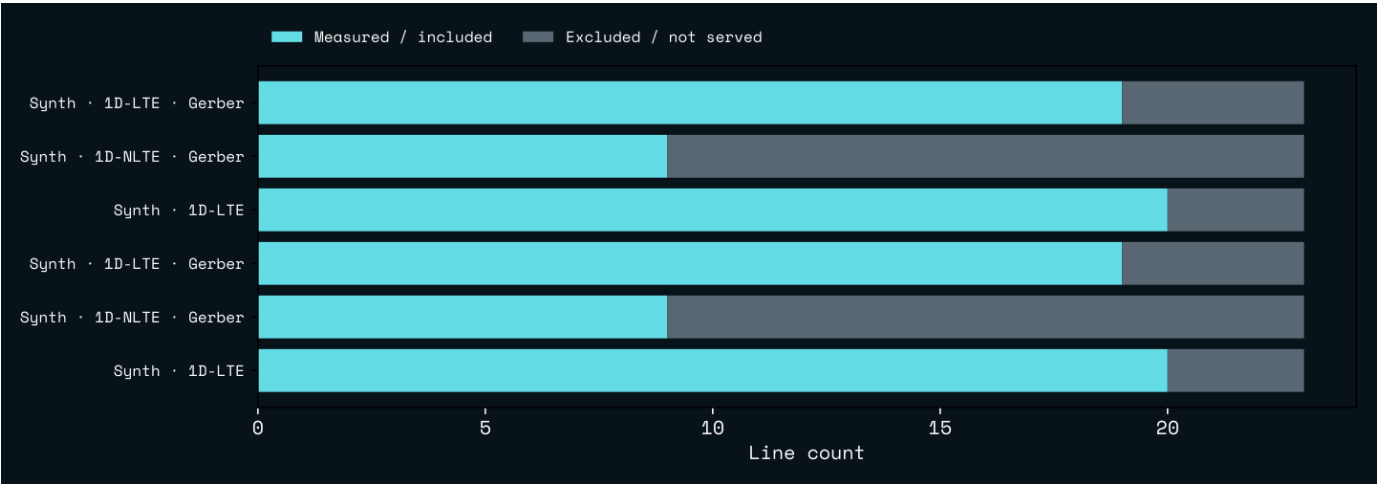
molecules OFF vs ON on THE SAME CODE and the same pool. Differencing against the published values instead would have measured the molecular lever PLUS every code change since the feed was written on Sept 3 -- which is about -0.035 dex of the total. [Source evidence](#)

HARPS arm offset



Current Fe II VIS products, separated by holding and engine. The arm offset persists after the curated CH artifact was removed. [Source evidence](#)

CRIRES+ H model coverage



Included and excluded line counts from each current product. Excluded rows are not automatically failed fits; the per-line reason distinguishes unavailable NLTE coverage. [Source evidence](#)

## Download the evidence

6686 graded per-line measurements exported. Evidence for 22 graded products is missing or does not reproduce the feed; 0 exported lines lack an unambiguous gf match. The download is explicitly incomplete until those upstream records are supplied.

[Graded per-line abundances \(CSV\)](#)

[All product fields \(CSV; nested fields preserved as JSON\)](#)

[Per-product line-evidence coverage \(CSV\)](#)

[Source Fe.json](#)

## Reproducibility

Generator scripts/generate\_fe\_publication.py v1.0.0

Science source commit 0e4539da63c8fe63facab98cae5166e98219126b

Fe.json v1.221 · feed timestamp 2026-09-19T05:36:46Z

Generated 2026-09-19T16:52:30+00:00

Feed SHA-256 66421362d81e61e8d3aa6a412e1ae362743c05e41fb8b12a2331347f890395b0

[Refreshed product tracker](#) · [Build manifest](#) · [Fe II appendix](#)

## References

Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). [The chemical make-up of the Sun: A 2020 vision](#) (doi: 10.1051/0004-6361/202140445). AGSS21 Solar reference and the green literature comparison; Table A.2 defines the Reference Grade Fe I line set.

Lodders, K.; Bergemann, M.; Palme, H. (2025). [Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites](#) (doi: 10.1007/s11214-025-01146-w). Table 6 present-day Solar composition, shown as the gold literature comparison.

Amarsi, A. M.; Liljegren, S.; Nissen, P. E. (2022). [3D non-LTE iron abundances in FG-type dwarfs](#) (doi: 10.1051/0004-6361/202244542). The Amarsi 3D non-LTE iron correction grid used by the displayed products.

Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). [Non-LTE radiative transfer with Turbospectrum](#) (doi: 10.1051/0004-6361/202243673). The Gerber Turbospectrum non-LTE synthesis implementation named in the model rows.

Ruffoni, M. P.; Pickering, J. C.; Allende Prieto, C.; Nave, G. (2013). [Infrared Laboratory Oscillator Strengths of Fe I in the H-Band](#) (doi: 10.1088/0004-637X/779/1/17). Laboratory transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.

Den Hartog, E. A.; Ruffoni, M. P.; Lawler, J. E.; Pickering, J. C.; Lind, K.; Brewer, N. R. (2014). [Fe I Oscillator Strengths for Transitions from High-lying Even-parity Levels](#) (doi: 10.1088/0067-0049/215/2/23). Laboratory transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.

Ruffoni, M. P.; Den Hartog, E. A.; Lawler, J. E.; Brewer, N. R.; Lind, K.; Nave, G.; Pickering, J. C. (2014). [Fe I oscillator strengths for the Gaia-ESO survey](#) (doi: 10.1093/mnras/stu780). Laboratory transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.

Belmonte, M. T.; Pickering, J. C.; Ruffoni, M. P.; Den Hartog, E. A.; Lawler, J. E.; Guzman, A.; Heiter, U. (2017). [Fe I Oscillator Strengths for Transitions from High-lying Odd-parity Levels](#) (doi: 10.3847/1538-4357/aa8cd3). Laboratory transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.